

Control Systems (EEE/ECE/INSTR F242)

Tutorial 5 (10 February 2026)

1. An armature-controlled DC motor and its speed-torque characteristics is shown in Fig. 1(a) and Fig. 1(b). Find the transfer function $\theta_L(s)/E_a(s)$.

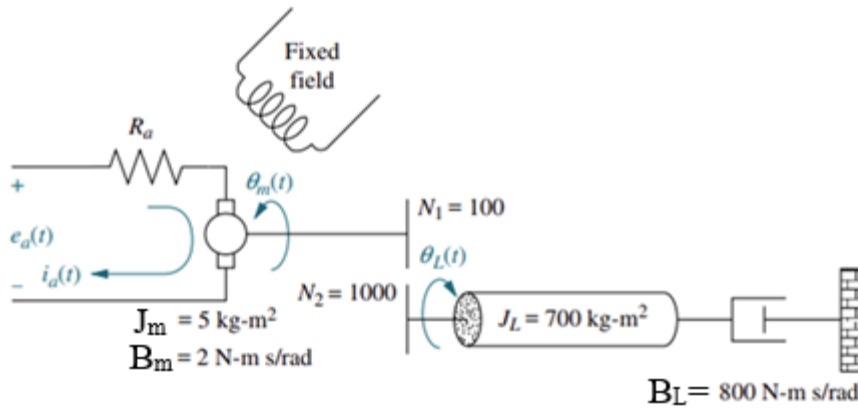


Fig. 1(a)

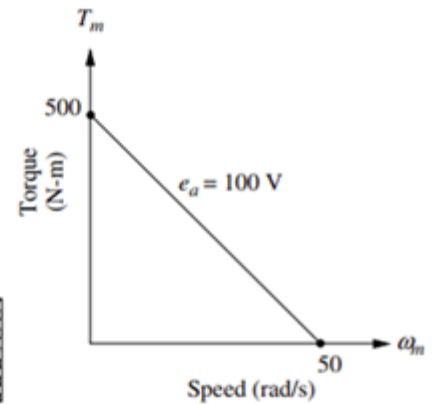


Fig. 1(b)

2. The schematic diagram of a servo motor is shown in Fig. 2(a). For this system, draw the block diagram and determine the transfer function $\theta_L(s)/\theta_R(s)$. The following parameters are given:

- Sensitivity of the synchro error detector (K_s) is 30 V/rad,
- Amplifier gain (K_A) is 20 V/V,
- Moment of inertia of load (J_L) = 1 kg-m²; Friction coefficient of load (B_L) = 4 Nm/rad/s.
- Moment of inertia and friction coefficient of motor are negligible.

The linearised torque-speed characteristic of the motor with 115 volts, 50 Hz applied to control winding is shown in Fig. 2(b). Take $\theta_L/\theta_m = 1/2$ and $\theta_m/\theta_s = 1/1$.

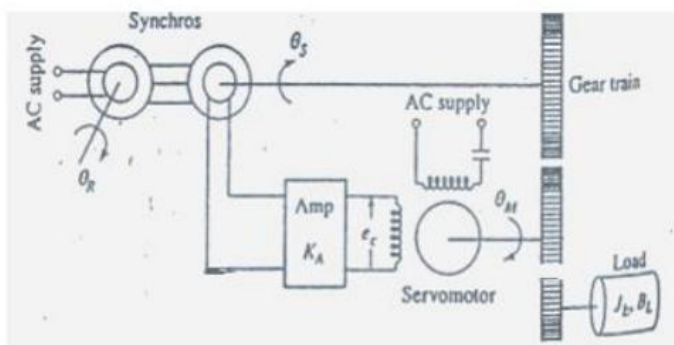


Fig. 2(a)

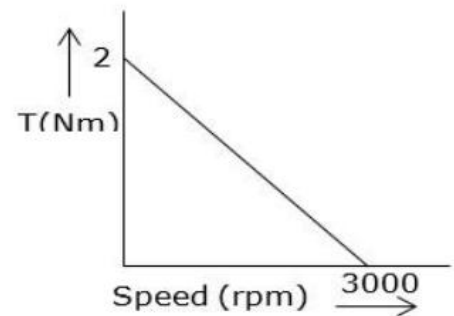


Fig. 2(b)
